

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

Nov/Dec 2012

EE206 – Digital Electronics & microprocessor

Date: 08/11/2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Explain laws of Boolean algebra. [05]
(b) Explain the Ex-OR gate. Specify its symbol and write its truth table. [02]
- Q - 2 (a) Implement the logic function of the below expression using 4:1 multiplexer. [06]
 $F(A,B,C) = \sum m(1,3,4,6)$
(b) Explain the working of a half adder? Draw its logic diagram. [04]
(c) Write notes on 2-bit synchronous up Counters. [04]

OR

- Q - 2 (a) Write notes on negative edge-triggered J-K flip-flop. [06]
(b) Distinguish between Demultiplexer and Decoder [04]
(c) Explain Serial In Parallel Out mode of shift register [04]
- Q - 3 (a) Prove that [06]
i) $(A+B'+AB)(A+B')(A'B) = 0$
ii) $AB'C + B + BD' + ABD' + A'C = B + C$
iii) $ABC + BC'D + A'BC = B(C+D)$
(b) Convert $(2003.31)_{10}$ into its equivalent hex number. [04]
(c) i) $AB + A'B' = D$ represents _____ gate. [04]
ii) Output of EX-NOR with inputs A and B is _____.
iii) The minterm corresponding to 011 combination is _____.
iv) The noise performance is _____ in TTL family than CMOS family devices.

OR

- Q - 3 (a) Minimize the following expression using K-map and realize using the basic gates. [06]
 $Y = A'B'C'D + A'B'CD' + AB'C'D + AB'CD' + AB'CD + ABCD' + ABCD$
(b) Convert the hex number $(68.4B)_{16}$ in to equivalent octal number. [04]

- (c) Minimize the following expression using K-map [04]
 $Y = \sum m(0,1,5,9,13,14,15) + d(3,4,7,10,11)$

SECTION – II

- Q - 4 (a) Write similarities and differences between the instructions CALL and RET & PUSH and POP instructions. [05]

- (b) Calculate the address line required for an 8K-byte memory chip with 8085 microprocessor [02]

- Q - 5 (a) Draw the complete pin diagram of 8085 microprocessor and explain the function of each pin in detail. [07]

- (b) What is flag? Draw the format of 8085 flag register and explain in detail. [07]

OR

- Q - 5 (a) Write 8085 microprocessor based assembly program for any three of the following. [09]
 Explain the program logic in brief.

- (i) Add 98H and 9AH. The 1st number 98H is in the memory location 2501H and 2nd number 9AH is in the memory location 2502H.

- (ii) Shift a 16-bit number 7596 H by one bit.

- (iii) To find the largest number in a data array. The number in a series are :

99, 76, 98 .

- (iv) Find the one's complement of an 8-bit number.

- (b) Write a program to count continuously in hexadecimal from FFH to 00H in a system with a 0.5 μ S clock period. Use register C to setup a one millisecond delay between each count and display the numbers at one of the output ports. [05]

- Q - 6 (a) Give the classification of memory and explain in brief. [07]

- (b) Explain the function of following instructions. [07]

- | | | |
|-----------|----------------|------------------------------|
| (i) ADD r | (ii) LDA addr. | (iii) XCHG |
| (i) CMA | (v) RRC | (vi) INR M (vii) MVI M,data. |

OR

- Q - 6 (a) Write a program to generate a continuous square wave with the period of 500 μ S. [07]
 Assume the system clock period is 325 ns, and use bit D₀ to output the square wave.

- (b) Write an assembly language program for multiplying 84H by 56 H. [07]
